Mumbai Delivery Route Optimizer

Results Report

Comparison of TSP and CVRP methods with a Monte Carlo simulation of delivery days

Submitted by

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1 Introduction and setup

This report summarises what the experiments in the Mumbai Route Optimizer project show. The project compares classic route-optimization methods for a delivery depot in Kurla, Mumbai, serving 45 stops from Colaba in the south to Borivali in the north, plus Mulund and Powai in the east.

All numbers come from `python run.py` with seed 42, Google OR-Tools enabled (10 s time limit per problem) and 2000 simulated days per method. The script was run three times; every distance and every simulation figure was identical across runs and only the runtimes changed, so the seeded results are reproducible.

Data and assumptions. Distances are straight-line (haversine) distances multiplied by a detour factor of 1.35, so all kilometre figures are estimates and not driven distances. The number of parcels per stop (1 to 10) is simulated, with a total demand of 277 parcels. The speeds, rain frequency, service times and failure rate used in the simulation are assumptions, not measurements.

2 Part 1: one vehicle, all 45 stops (TSP)

Method	Distance (km)	Better than random	Runtime (s)
Random order (avg. of 500)	715.64	–	0.00
Nearest neighbour	190.55	73.4%	0.00
Nearest neighbour + 2-opt	150.74	78.9%	0.01
Genetic algorithm	148.76	79.2%	15–19
Google OR-Tools	**147.61**	79.4%	≈10

Table 1: TSP results. Runtimes of the genetic algorithm and OR-Tools vary a little between runs.

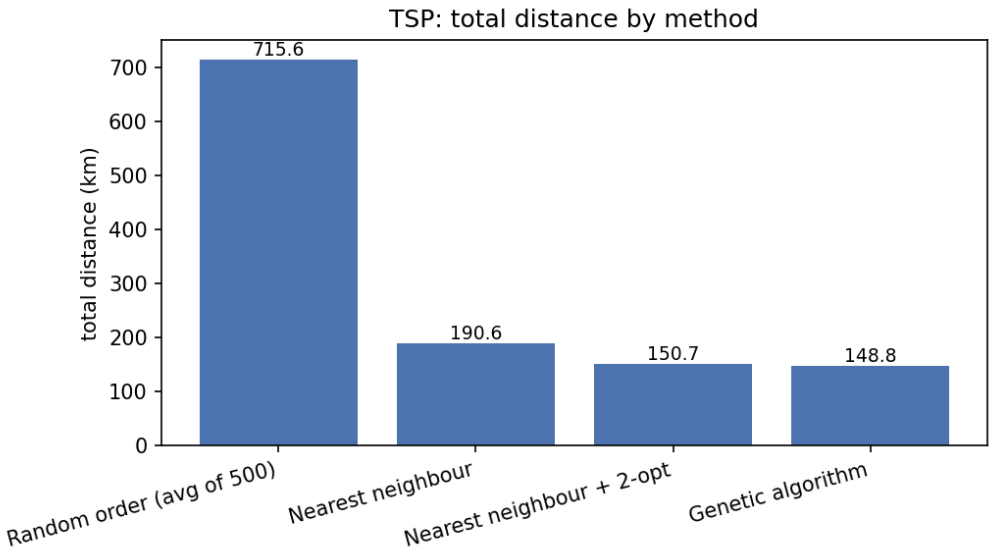


Figure 1: TSP tour length by method.

Observations

- **Any sensible method beats random order by a large margin.** Plain nearest neighbour already reduces the tour from 715.6 km to 190.6 km (−73%).
- **2-opt is the best value.** It removes a further 20.9% (190.55 to 150.74 km) in about 0.01 s.
- **The last few percent are expensive.** The genetic algorithm is only 1.3% shorter than NN + 2-opt but takes roughly 1500 times longer. OR-Tools is 2.1% shorter than NN + 2-opt and 0.8% shorter than the genetic algorithm, in about 10 s.
- **The genetic algorithm converges early.** In Figure 2 the best tour passes the NN + 2-opt level at around generation 100 and then flattens, with only two small improvements near generations 600 and 980. One thousand generations is more than this instance needs.

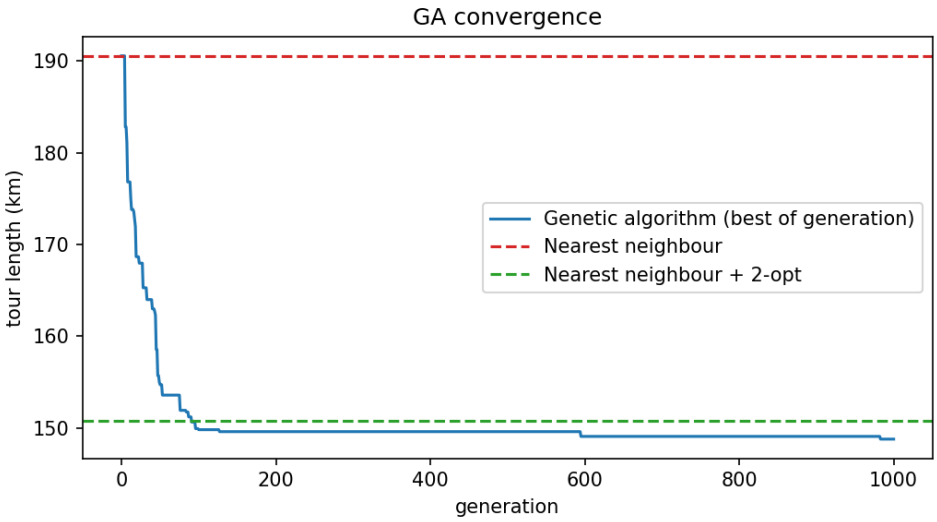


Figure 2: Convergence of the genetic algorithm against the NN and NN + 2-opt tour lengths.

Conclusion. OR-Tools gives the shortest tour (147.61 km), but NN + 2-opt is within 2.1% of it at almost no cost. The three best methods lie within about 2% of each other, which suggests that the optimum for this instance is close to 147 km (an observation, not a proof).

3 Part 2: four vehicles, capacity 75 (CVRP)

The fleet capacity is 300 parcels for a demand of 277, so the fleet is about 92% full.

Method	Distance (km)	Vehicles	Better than greedy	Runtime (s)
Greedy capacity-aware NN	276.64	4	–	0.00
Greedy + 2-opt + relocate	240.40	4	13.1%	0.00
Sweep + 2-opt + relocate	210.33	4	24.0%	0.01
Google OR-Tools	**190.80**	4	31.0%	≈10

Table 2: CVRP results for 4 vehicles of capacity 75.

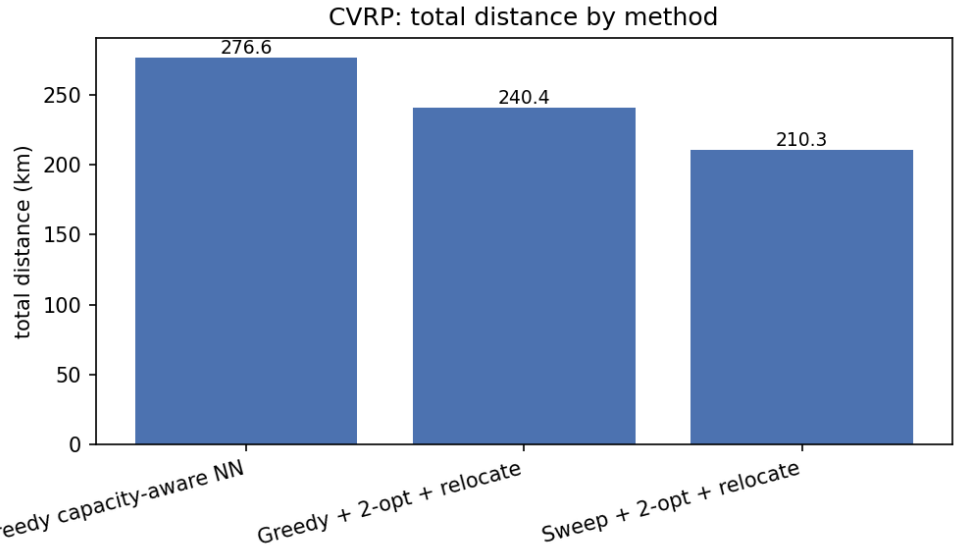


Figure 3: Total distance of the CVRP plans by method.

Observations

- **The starting solution matters a lot.** Local search improves both starting points, but the sweep start ends 12.5% shorter than the greedy start (210.3 vs. 240.4 km). Sorting stops by angle around the depot gives compact, non-overlapping routes (Figure 4).
- **OR-Tools is 9.3% shorter than the best heuristic** (190.8 vs. 210.3 km) and 31.0% shorter than greedy. Sweep + local search closes most of the gap, but not all of it.
- **The OR-Tools routes are unbalanced in distance** (Table 3). Vehicles 1 and 2 drive about 55–58 km, vehicles 3 and 4 only about 38–40 km. The solver minimises total distance, not the longest route, so nothing pushes it to balance the vehicles. This matters in Part 3, because the day ends when the slowest vehicle returns.
- **Capacity is almost binding.** Vehicle 2 is exactly full and vehicle 3 is at 74/75.

Vehicle	Stops	Load	Distance (km)
1	11	70/75	57.7
2	14	75/75	55.4
3	12	74/75	37.6
4	8	58/75	40.1

Table 3: Best CVRP plan found by OR-Tools.

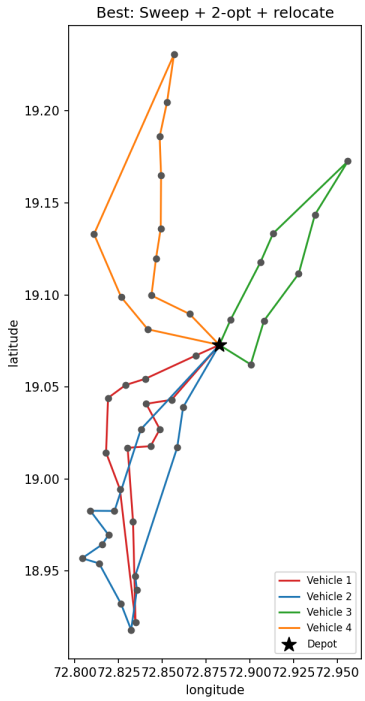


Figure 4: Routes of the Sweep + 2-opt + relocate plan (4 vehicles, star = depot).

Conclusion. OR-Tools gives the lowest total distance (190.8 km). Where a simple method without extra dependencies is preferred, sweep + 2-opt + relocate is a reasonable second choice at 10.2% more distance than OR-Tools.

4 Part 3: Monte Carlo simulation

Each plan was simulated for 2000 random days. The assumptions are 25 km/h base speed, rush-hour slowdowns, rain on 15% of days, random service times, 5% failed deliveries and a 5.0 h shift. The "finish" time is the time at which the last vehicle is back at the depot.

Method	Plan km	Route h	Finish h	P95 h	Overtime	OT h	Cost INR
Greedy NN	276.6	4.94	6.15	7.77	96.6%	1.91	9,797
Greedy + 2-opt + relocate	240.4	4.54	5.54	6.92	78.2%	0.84	9,095
Sweep + 2-opt + relocate	210.3	4.16	4.65	5.76	21.1%	0.19	8,572
Google OR-Tools	190.8	3.97	4.65	5.88	22.1%	0.19	**8,336**

Table 4: Simulation results. Route h = mean route time, Finish h = mean finish time, P95 h = 95th percentile of the finish time, Overtime = probability of exceeding the 5 h shift, OT h = mean overtime.

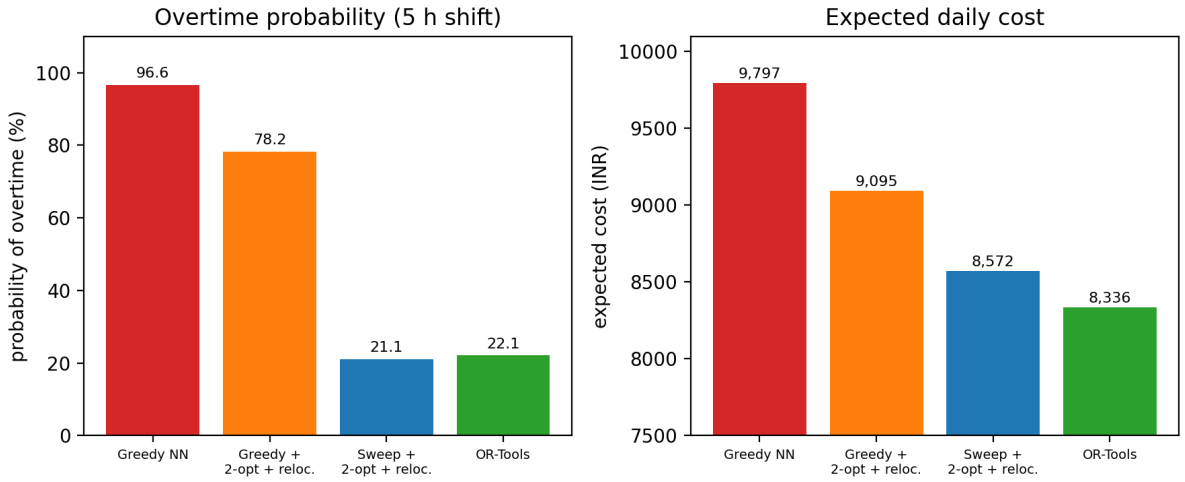


Figure 5: Overtime probability and expected cost by method (drawn from the values in Table 4).

Observations

- **Plan quality turns into operational risk.** A greedy plan ends after the 5 h shift on about 97% of simulated days (mean finish 6.15 h, P95 7.77 h). The sweep and OR-Tools plans reduce this to about 21–22%.
- **A shorter plan helps less than the planned distance suggests.** OR-Tools plans 9.3% fewer kilometres than Sweep, yet both have the same mean finish time (4.65 h) and almost the same overtime probability (22.1% vs. 21.1%). Sweep even has a slightly better P95 (5.76 vs. 5.88 h). The likely reason is the route imbalance seen in Part 2: the finish time is set by the longest route, and OR-Tools minimises total distance. This is an interpretation and was not tested directly.
- **OR-Tools still has the lowest expected cost** (8,336 INR), 14.9% below greedy and 2.8% below Sweep, so it remains the best choice on cost, although the margin over Sweep is small.
- **The ordering of the methods is stable.** Ranking by distance and by expected cost agree: greedy $<$ greedy + local search $<$ sweep $\approx$ OR-Tools.

5 Fleet size what-if

The sweep + 2-opt + relocate plan was re-run for 3 to 7 vehicles. The capacity per vehicle is scaled so that the total fleet capacity is about 110% of the demand (for example 3 $\times$ 102 or 7 $\times$ 44). The 4-vehicle row therefore uses capacity 77 instead of 75, but gives the same 210.3 km plan as Part 2.

Vehicles	Capacity	Plan km	P95 finish h	Overtime	Cost INR
3	102	197.2	7.87	97.8%	7,224
4	77	210.3	5.76	21.1%	8,572
5	61	244.0	5.91	23.4%	10,473
6	51	251.9	5.44	13.5%	12,038
7	44	279.3	4.79	2.7%	13,855

Table 5: Fleet size what-if for the sweep + 2-opt + relocate plan.

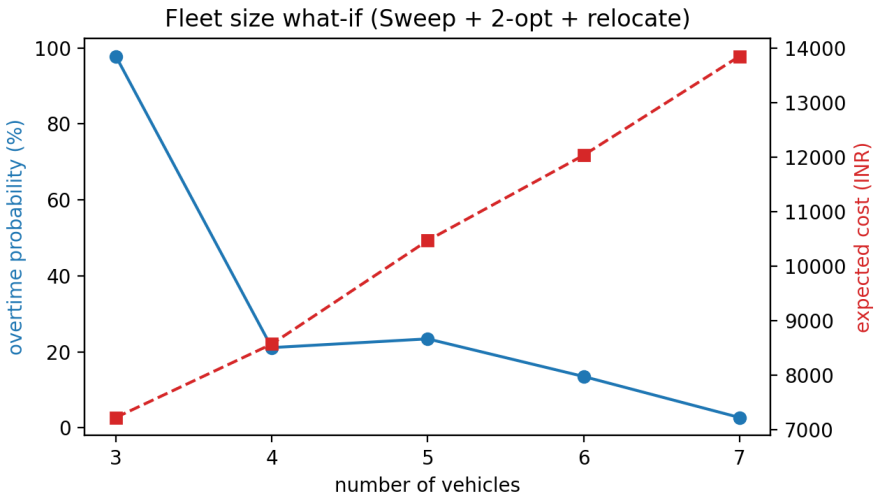


Figure 6: Overtime probability and expected cost against fleet size (drawn from Table 5).

Observations

- **Three vehicles are not feasible for a 5 h shift.** Overtime occurs on 97.8% of days, even though it is the cheapest plan on paper.
- **Cost rises with every extra vehicle** (roughly 1,300 to 1,900 INR per vehicle). Planned kilometres also grow, because each vehicle makes its own trip out of and back to the Kurla depot.
- **Overtime falls with more vehicles, but not smoothly.** Six vehicles reduce it to 13.5% and seven vehicles to 2.7%. Reaching 2.7% costs about 5,300 INR more per day than the 4-vehicle plan (13,855 vs. 8,572).
- **The 5-vehicle row is worse than the 4-vehicle row** (P95 5.91 vs. 5.76 h, overtime 23.4% vs. 21.1%), which should not happen with an extra vehicle. The most probable explanation is that the sweep heuristic produced an uneven split at that fleet size and left one long route. This is an artifact of the heuristic rather than a real effect, but it was not checked route by route.

Conclusion. There is no free option. Four vehicles is the cheapest feasible plan but runs over about one day in five. Seven vehicles nearly removes overtime but raises the cost by about 62%. The right choice depends on what an overtime day costs in practice.

6 Overall conclusions

1. **Start with a good construction heuristic, then apply local search.** Sweep + 2-opt + relocate gets within about 10% of OR-Tools on the CVRP, and NN + 2-opt within about 2% on the TSP, in hundredths of a second.
2. **OR-Tools gives the best plans in every comparison**, on distance and on expected cost, with a 10 s time limit.
3. **The genetic algorithm is not worth it here.** It is slower than OR-Tools and barely better than 2-opt.
4. **Minimum distance is not minimum risk.** Once randomness is added, the OR-Tools advantage shrinks. A formulation that balances route lengths, or minimises the longest route, is the obvious next experiment.

5. **The 5 h shift is the binding constraint**, not vehicle capacity. Fleet size should be decided from the overtime and cost trade-off in the what-if table.

7 Suggested next steps

- Minimise the longest route, or add a balance penalty, in OR-Tools and re-run the simulation.
- Inspect the 5-vehicle sweep solution to confirm the uneven-split explanation.
- Repeat the study with real OSRM road distances and with several seeds to obtain confidence intervals.
- Test larger instances (100+ stops), where OR-Tools and the simple heuristics are likely to diverge more.